

Ehrenfest theorem

The Ehrenfest theorem tells us expectation values obey the laws of classical mechanics. For example, by the classical momentum formula $p = mv$ we have

$$\langle p \rangle = m \frac{d}{dt} \langle x \rangle \quad (1)$$

To verify (1), start with momentum expectation value $\langle p \rangle$.

$$\langle p \rangle = \int \psi^* p \psi \, dx$$

By the substitution

$$p = \frac{im}{\hbar} [H, x]$$

we have

$$\langle p \rangle = \frac{im}{\hbar} \int \psi^* [H, x] \psi \, dx$$

Expand the commutator.

$$\langle p \rangle = \frac{im}{\hbar} \int \psi^* H x \psi \, dx - \frac{im}{\hbar} \int \psi^* x H \psi \, dx$$

By the Schrodinger equation we have the substitutions

$$H\psi = i\hbar \frac{d}{dt} \psi$$

and

$$\psi^* H = (H\psi)^* = -i\hbar \frac{d}{dt} \psi^*$$

Hence

$$\langle p \rangle = \frac{im}{\hbar} \int -i\hbar \frac{d}{dt} \psi^* x \psi \, dx - \frac{im}{\hbar} \int \psi^* x i\hbar \frac{d}{dt} \psi \, dx$$

Cancel $\hbar$.

$$\langle p \rangle = m \int \frac{d}{dt} \psi^* x \psi \, dx + m \int \psi^* x \frac{d}{dt} \psi \, dx$$

By the product rule for differentiation

$$\langle p \rangle = m \frac{d}{dt} \int \psi^* x \psi \, dx$$

Hence

$$\langle p \rangle = m \frac{d}{dt} \langle x \rangle$$

Eigenmath script